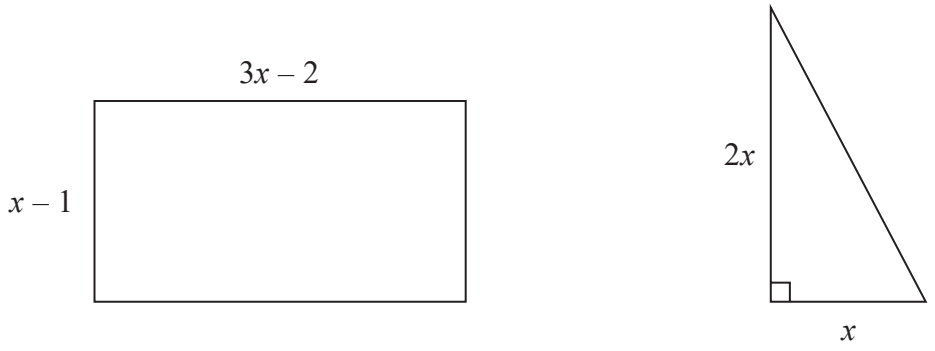
23 Here is a rectangle and a right-angled triangle.



All measurements are in centimetres.
The area of the rectangle is greater than the area of the triangle.

Find the set of possible values of x.

Rectangle: $(x - 1)(3x - 2)$. Triangle: $\frac{1}{2} \times x \times 2x = x^2$.

1 mark
Form both area expressions

×	$3x$	-2
x	$3x^2$	$-2x$
-1	$-3x$	2

$3x^2 - 2x - 3x + 2 > x^2$

Subtract x^2: $2x^2 - 5x + 2 > 0$.

1 mark
Rearrange into a quadratic inequality

×	$2x$	-1
x	$2x^2$	$-x$
-2	$-4x$	2

Split $-5x$ into $-x - 4x$.

$(2x - 1)(x - 2) > 0$

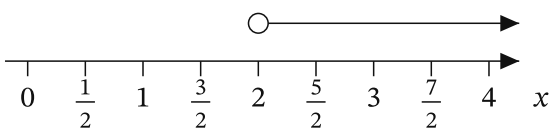
1 mark
Factorise to find the critical values

Critical values: $x = \frac{1}{2}$ and $x = 2$.

1 mark
Find both boundary values

The product is positive when $x < \frac{1}{2}$ or $x > 2$.

But the length $x - 1$ must be positive, so $x > 1$.



1 mark
Keep only the geometrically valid branch and exclude equality

$x > 2$

(Total for Question 23 is 5 marks)